TAMOXIFEN

VTE & endometrial cancer
Avoid CYP2D6 inhibitors

CHEMICAL NAME (Z)-2-[4-(1,2-diphenylbut-1-enyl)phenoxy]-N,N-dimethylethanamine (citrate)
CLASS Selective oestrogen receptor modulator (SERM) — hormonal anticancer
TRADE NAMES Nolvadex · Soltamox (liquid) · Tamofen · Istubal

THERAPEUTIC USES / INDICATIONS

- Oestrogen receptor-positive breast cancer (premenopausal and postmenopausal) — adjuvant (5–10 years) and metastatic.
- Risk reduction in women at high familial risk; DCIS.
- Male breast cancer; gynaecomastia (off-label); anovulatory infertility (some countries).

ADMINISTRATION

- **PO** once daily at the same time, with or without food.
- Non-hormonal contraception needed in premenopausal women.

DOSAGES

- **Breast cancer:** 20 mg once daily for 5–10 years.
- **Risk reduction:** 20 mg daily for 5 years.

PHARMACOKINETICS

- **Peak:** 4–7 h. Steady state 4 weeks.
- Prodrug activated by CYP2D6 to endoxifen (main active metabolite).
- Excreted in faeces. **t½:** 5–7 days (metabolite 14 days).

NURSING MANAGEMENT

- Check concomitant antidepressants (avoid paroxetine/fluoxetine).
- Assess VTE risk; stop 4–6 weeks before major elective surgery (per guidance).
- Ask about abnormal vaginal bleeding, visual changes.
- Support adherence over years.

PHARMACODYNAMICS

- **Antagonist** at oestrogen receptors in breast tissue → ↓ tumour growth.
- **Agonist** in endometrium (hyperplasia, cancer), bone (protects in postmenopausal) and liver (↑ clotting factors).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hot flushes, night sweats.
- Vaginal discharge, dryness, irregular periods.
- Nausea, fluid retention, mood changes.

ADVERSE EFFECTS (SERIOUS)

- Venous thromboembolism (DVT, PE), stroke.
- Endometrial hyperplasia and cancer (postmenopausal) — any abnormal vaginal bleeding needs investigation.
- Cataracts, retinopathy.
- Hepatotoxicity (fatty liver); teratogenicity.

CONTRAINDICATIONS & CAUTIONS

- Pregnancy, breastfeeding.
- History of VTE (relative; specialist decision).
- Use with anastrozole/letrozole (antagonism).

KEY INTERACTIONS

- CYP2D6 inhibitors (paroxetine, fluoxetine, bupropion); ↓ endoxifen → reduced efficacy; use venlafaxine for hot flushes.
- **Warfarin:** ↑ INR (avoid/monitor).
- **Aromatase inhibitors:** do not combine.

PATIENT EDUCATION

- Take every day for the full 5–10 years.
- Report any vaginal bleeding after menopause, leg pain/swelling, chest pain, breathlessness, visual changes.
- Use non-hormonal contraception (condoms, copper IUD).
- Tell your doctor before starting antidepressants.

KEY POINTS

- SERM — blocks oestrogen in breast.
- VTE and endometrial cancer risk.
- 20 mg daily for 5–10 years.
- Activated by CYP2D6; avoid paroxetine/fluoxetine.